Héctor Ochando

SOFTWARE ENGINEER

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in [Hector Ochando]()

ABOUT

I am a Software Engineer with a recent degree in Computer Science from Sheffield Hallam University, specializing in performance-critical development and engine architecture. I have demonstrated the ability to create optimized systems through custom engine projects, including real-time rendering pipelines using modern techniques and memory management frameworks. With a strong foundation in low-level programming and modern development practices, I approach problems with a practical and methodical mindset, always eager to dive deep into the code to understand the underlying structure and learn from them. While my primary focus is on engine development, rendering, and performance optimization, I am also open to roles in tools programming or gameplay programming, where technical expertise and creative problem-solving can drive impactful results.

RELEVANT EXPERIENCE

Tiny Terrors: Junior *Programmer*

SEPTEMBER 2023 - JULY 2024, VALENCIA, ALREADY IN **STEAM**

Developing Shelley Manor game with Unreal Engine 5 and C++.
Developed a modular camera system and other gameplay features.
Work focused on artificial intelligence.

EDUCATION

Sheffield Hallam University, Computer Science for Games, BSC HON

SEPTEMBER 2024 - MAY 2025, SHEFFIELD

I studied the final year, where I have been able to learn about game development for **PS5** and graphics techniques such as **Ray Tracing** with **DX12[DXR]**.

ESAT, HND in Computing, BTEC Level 5

OCTOBER 2021 - JULY 2024, VALENCIA

I studied for three years at ESAT, where I became proficient in C and C++. In addition, I gained knowledge in ARM Assembly, OpenGL and DirectX 11. As well as developing and publishing a game as a final project in Unreal Engine.

SKILLS

C++	C	C#
Unreal Engine	Unity	
HLSL	GLSL	PSSL
DirectX 11/12	OpenGL	DXR
Git	Perforce	
Render Doc	Nsight	PIX
Razor GPU		
Visual Studio	CMake	Premake

LANGUAGES

Spanish: Native Proficiency

English: IELTS 6.0 (2024)

Catalan: Native Proficiency

AWARDS

Top 5 – Best Hobby Game
Game Development World Championship 2024 (Winter Season)
Awarded for the game **Shelley Manor**, published on [Steam]().

Finalist – Best Student Game
Gamescom Latam 2025
Finalist recognition for **Shelley Manor**

Finalist – Best Animation for Video Games
Quirino Awards 2025
Recognized for outstanding animation in **Shelley Manor**.

STUDENT/PERSONAL PROJECTS

PS5 Game Engine: *Graphics Programming & Engine Systems Programming*

Developed a custom engine using the **PlayStation 5 SDK**, gaining hands-on experience with the console's development environment — not only the graphics API**[AGC]**, but also its supporting tools**[Razor GPU]** and workflows. Throughout the project, I also learned about the packaging process for PS5, including asset management, build configuration, and deployment. This experience helped me deepen my understanding of engine and graphics systems, and continue expanding my skills in console-specific development. More info [here]().

DirectX12 Ray Tracing: *Graphics Programming*

Developed a fully raytraced demo in **C++** and **DirectX12 [DXR]**, where I learned how ray tracing works and how to use it. Also its strengths and weaknesses, learning more about its use in hybrid graphics pipelines. More info here.

Everfrost Engine: *Graphics Programming & Engine Systems Programming*

Everfrost Engine is a work-in-progress project focused on modern graphics and engine systems programming. It explores techniques such as **Visibility Buffer** rendering, **RHI**-based API abstraction (**DX12** now), bindless resources, and **memory footprint optimizations**, with integration of the **Optick profiler** as an additional tool for performance analysis. More info here.

Andromeda Engine: *Graphics Engine*

Development of an engine written in **C++** with **DirectX 11** and **OpenGL** as rendering backends. I work with APIs like PhysX and OpenAL. This experience has allowed me to learn about the basics of graphics, as well as the basic systems of a videogame powerhouse, going deep into graphics, **ECS** and **multithreading.** Using tools such as **Render Doc**, **PIX** or the Visual Studio debugger. More info here.

Memory Manager: *Low Level Programming*

Memory management with a custom Memory Manager.

Developed a custom **Memory Manager**, following the arena allocator pattern which I programmed in C to manage dynamic memory allocation and catch any memory leak. More info here.

Several ASM Demos: *Low Level Programming*

Voxel Ray Tracer terrain and Landscape terrain based on a height map. Programmed in C and ASM with an SDL. More info here.

Particle System: *Multithreaded Programming*

Built a **C++ multithreaded** particle system where particles update life, position, and collisions in real time using a **task-based architecture**, with performance measured and analyzed using the **Tracy profiler**. More info here.

ADTs: *Low Level Programming*

Developed an API written in **C / C++** composed by the following ADT: Memory Node, Vector, List, Double Linked List, Movable Head Vector, Circular Vector, Circular List, Circular Double Linked List, Queue, Stack. API severely influenced by the C++ STL. More info here.

Multi Agent System: *IA Demo*

Developed a demo of a multi agent system with different agent roles. Working with Unity and C#. More info here.